Department of CSE

Mini Project Report - Task 3

"Malware Family Classification Using CNN Baselines and an Improved GCN on the MaleBin Dataset"

Course Code: ICE 478/CSE475

Section: 2

Track: Track 2 - GCN (Image-to-Graph)

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1. Introduction

Task 3 completes the Track 2 image-to-graph pipeline for malware family classification on the MaleBin dataset. The objective was to improve the first PatchGCN from Task 2, measure the effect of graph design choices through ablation, compare the final GCN with the strongest CNN baseline, evaluate stability through five-fold stratified cross-validation, test statistical significance, and examine model decisions using complementary explainability methods.

The first PatchGCN used six handcrafted statistical features for each image patch and achieved a macro-F1 of 0.3771. In Task 3, these node attributes were replaced by learned CNN feature vectors before graph convolution. This change produced a much stronger graph representation while retaining a compact model.

2. Task 3 Objectives

- Improve the first GCN using learned image features and graph-model tuning.
- Perform ablation experiments on graph connectivity and batch normalization.
- Compare the final GCN with the Task 2 first GCN and the strongest CNN baseline.
- Report full multiclass metrics, ROC-AUC, precision-recall behavior, model size, and inference time.
- Run five-fold stratified cross-validation and statistical significance testing.
- Analyze one correct and one incorrect prediction with Grad-CAM, patch-level local surrogate importance, and graph-node occlusion importance.

3. Dataset Preparation and Data Integrity

The Task 3 notebook scanned 12,464 original grayscale malware images. SHA-256 hashes were computed before splitting so exact duplicates could be controlled. Cross-class duplicate groups were removed, within-class exact duplicates were reduced to one copy, and classes with fewer than 10 clean samples were excluded. The resulting experiment used 9,230 images across 35 malware families.

Item	Value
Original images	12,464
Cross-class duplicate groups removed	499
Usable images	9,230
Retained classes	35
Training	6,461 (70%)
Validation	1,384 (15%)
Test	1,385 (15%)
Leakage check	PASS

Class imbalance was handled with a WeightedRandomSampler during training. The validation and test distributions were kept unchanged. CNN inputs used 224 x 224 resolution, while graph-model inputs used 128 x 128 resolution.

4. Improved Image-to-Graph Model

The final model first uses a compact CNN backbone to learn local texture features. Adaptive pooling converts the feature map to an 8 x 8 grid. Each grid position becomes one graph node, producing 64 nodes per image with 128 learned features per node. The selected graph connects each node to its horizontal and vertical spatial neighbors and includes self-loops.

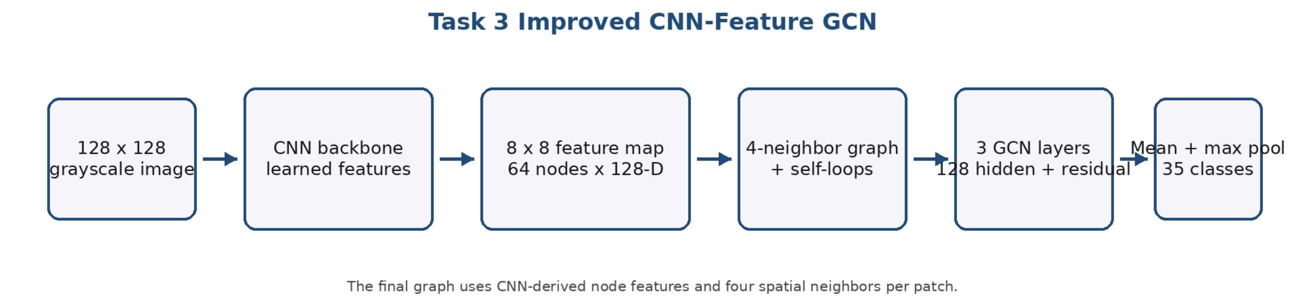

Figure 4.1: Improved CNN-feature GCN architecture

Three graph-convolution layers with 128 hidden features perform message passing. Batch normalization and ReLU are applied after graph convolution, dropout provides regularization, and a residual connection combines the last graph representations. Mean and maximum graph pooling are concatenated before the final 35-class classifier.

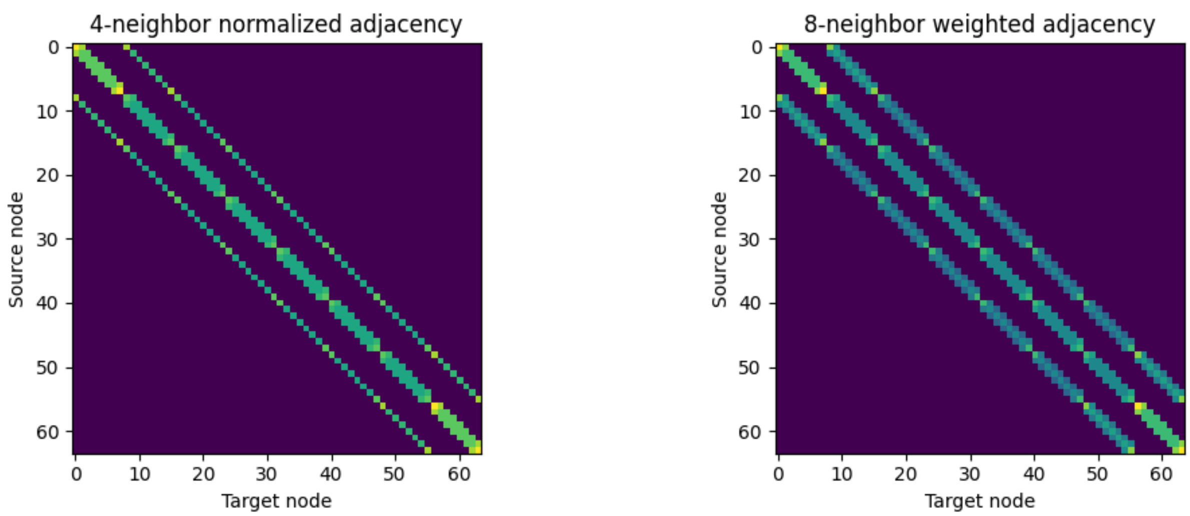

Figure 4.2: Normalized 4-neighbor and weighted 8-neighbor graph structures evaluated in Task 3

4.1 Training Settings

Setting	Value
Optimizer	AdamW
Learning rate	0.001
Weight decay	0.0001
Hidden width	128
Dropout	0.30
GCN depth	3
Fixed GCN epochs	5
Fixed ResNet18 epochs	6
CV epochs per fold	3
Scheduler	ReduceLROnPlateau
Early stopping	Enabled
Gradient clipping	1.0

5. Ablation Study

Four Task 3 variants were evaluated on the same held-out test split. The CNN-only model tests whether graph message passing adds value. The two GCN connectivity variants compare local 4-neighbor connections with a denser weighted 8-neighbor graph, and a no-BatchNorm variant measures the contribution of normalization.

Model / Ablation	Accuracy	Balanced Acc.	Macro-F1	ROC-AUC	Macro AP
CNN only - graph off	0.7487	0.7571	0.7191	0.9737	0.7873
CNN + GCN, 4-neighbor	0.7884	0.8076	0.7857	0.9789	0.8316
CNN + GCN, weighted 8-neighbor	0.7762	0.7852	0.7645	0.9804	0.8211
CNN + GCN, 8-neighbor, no BN	0.7285	0.7554	0.7199	0.9661	0.7738

The 4-neighbor model produced the strongest GCN macro-F1 of 0.7857. Compared with the same CNN backbone without graph message passing, macro-F1 increased by 0.0666. The weighted 8-neighbor graph was slightly weaker, so additional diagonal connections did not improve the selected evaluation metric. Removing batch normalization from the 8-neighbor model reduced macro-F1 to 0.7199.

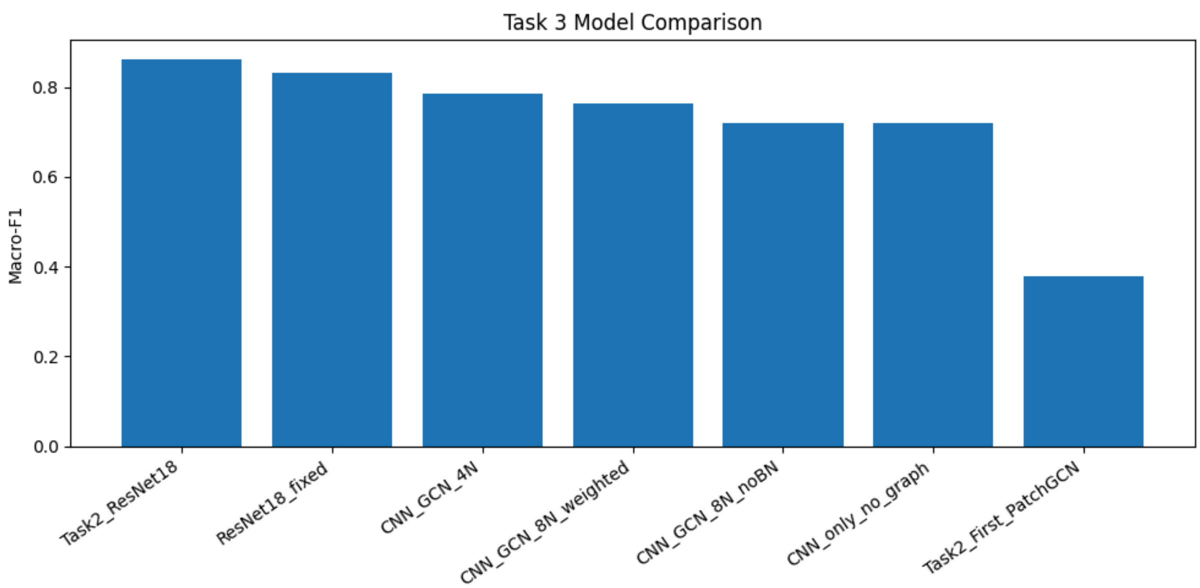

Figure 5.1: Macro-F1 comparison across Task 2 and Task 3 models

6. Final Model Performance

Model	Accuracy	Balanced Acc.	Macro-F1	Weighted-F1	Model Size	Inference
Task 2 ResNet18	0.8664	0.8811	0.8615	0.8648	-	-
Task 2 first PatchGCN	0.4433	0.5080	0.3771	0.3887	-	-
Task 3 ResNet18 paired run	0.8347	0.8519	0.8323	0.8365	42.76 MB	1.39 ms/image
Task 3 final CNN_GCN_4N	0.7884	0.8076	0.7857	0.7857	0.61 MB	1.20 ms/image

The improved GCN increased macro-F1 from 0.3771 for the first PatchGCN to 0.7857, an absolute increase of 0.4086. The paired Task 3 ResNet18 remained stronger on the fixed test set, but the final GCN became substantially more competitive while using a much smaller checkpoint.

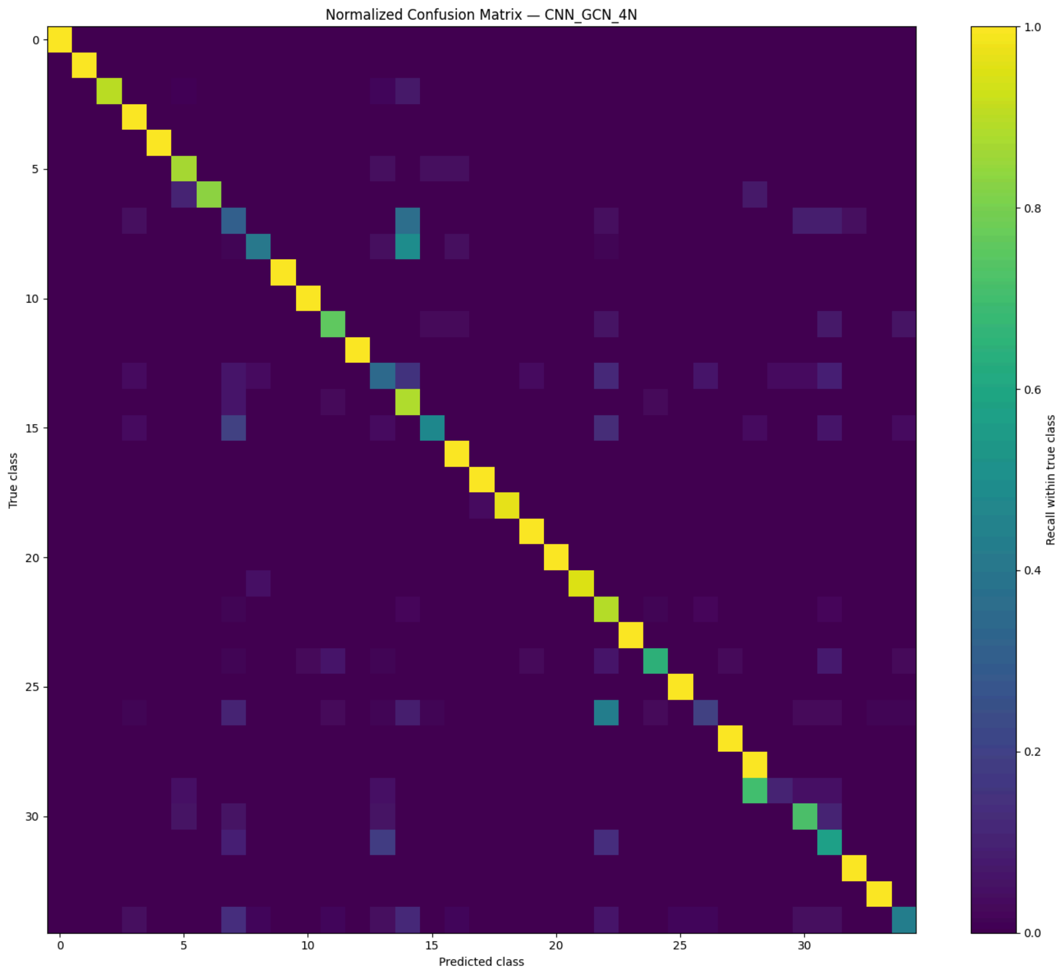

Figure 6.1: Normalized confusion matrix of the final CNN_GCN_4N model

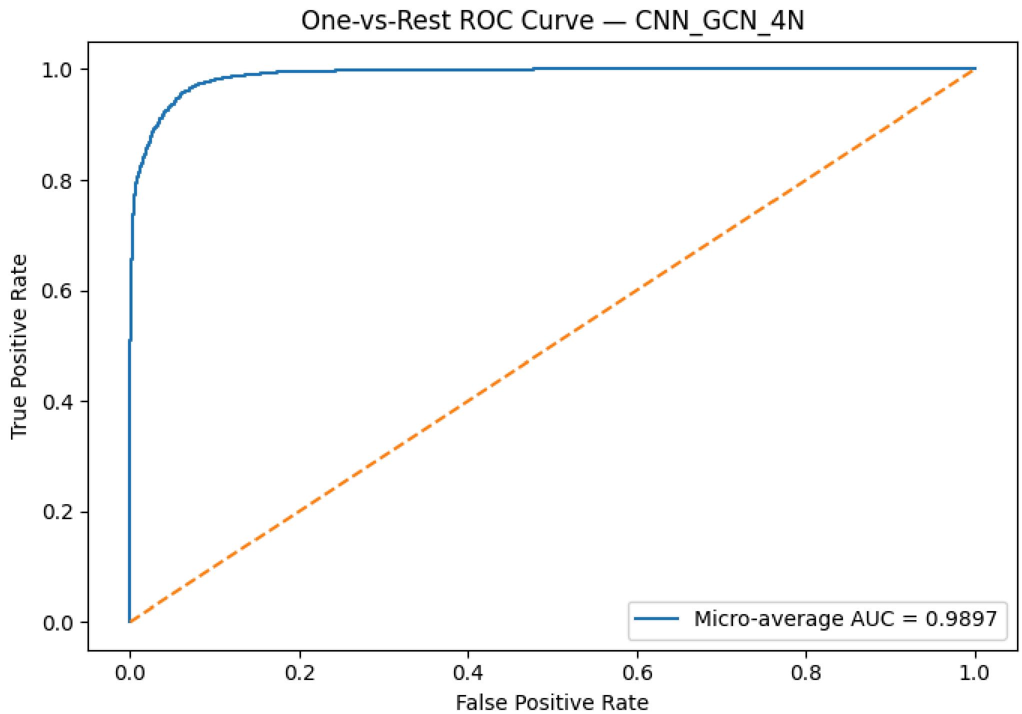

Figure 6.2: One-vs-rest ROC curve for the final GCN

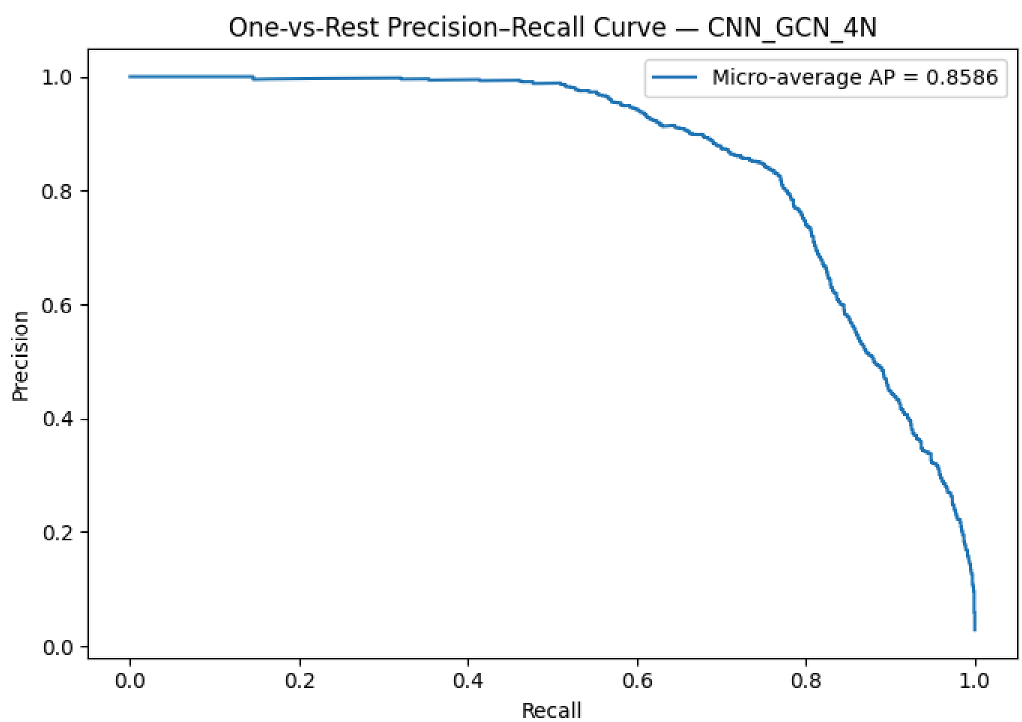

Figure 6.3: One-vs-rest precision-recall curve for the final GCN

The final GCN achieved macro one-vs-rest ROC-AUC = 0.9789 and macro average precision = 0.8316. The curves show strong ranking ability across the multiclass problem even though several minority families remain difficult.

7. Five-Fold Cross-Validation

Five-fold stratified cross-validation was run with the same fold assignments for ResNet18 and the selected CNN_GCN_4N architecture. A reduced three-epoch schedule per fold was used to keep the full comparison computationally practical.

Fold	ResNet18 Macro-F1	Final GCN Macro-F1
1	0.6507	0.6673
2	0.7358	0.6748
3	0.7684	0.6119
4	0.7768	0.6986
5	0.7230	0.7617

Metric	ResNet18 Mean +/- SD	Final GCN Mean +/- SD
Accuracy	0.7278 +/- 0.0735	0.7304 +/- 0.0295
Balanced accuracy	0.7654 +/- 0.0499	0.7289 +/- 0.0437
Macro-F1	0.7309 +/- 0.0501	0.6829 +/- 0.0543
Weighted-F1	0.7212 +/- 0.0741	0.7103 +/- 0.0347
Macro ROC-AUC	0.9782 +/- 0.0047	0.9725 +/- 0.0068
Macro average precision	0.8249 +/- 0.0165	0.7862 +/- 0.0326

The final GCN produced slightly higher mean accuracy and lower variation in accuracy, but ResNet18 achieved higher mean balanced accuracy, macro-F1, ROC-AUC, and average precision. The cross-validation results therefore do not support a claim that the final GCN consistently outperforms the CNN baseline.

8. Statistical Significance

Test	Comparison	Statistic / Counts	p-value	Conclusion
Exact McNemar	Task 3 ResNet18 vs final GCN	b = 128, c = 64	0.000004	Significant
Wilcoxon signed-rank	5-fold Macro-F1	W = 3.0	0.3125	Not significant

McNemar testing on the shared 1,385-sample test set found a significant paired difference between ResNet18 and the final GCN. In contrast, the Wilcoxon signed-rank test across the five paired fold-level macro-F1 values was not significant at alpha = 0.05. The statistical evidence therefore supports the fixed-test advantage of ResNet18 but does not establish a reliable fold-wise difference from only five folds.

9. Explainability Analysis

Explainability was examined with three complementary views: Grad-CAM for the ResNet18 baseline, patch-level local surrogate importance for the final GCN, and graph-node occlusion importance. One highly confident correct prediction and one highly confident error were selected automatically from the test set.

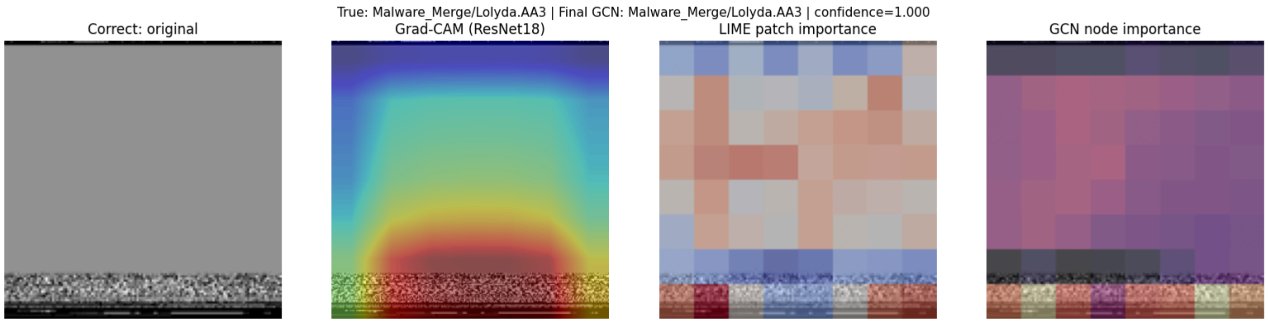

Figure 9.1: Correct prediction - Lolyda.AA3 classified as Lolyda.AA3 (GCN confidence 0.9998)

For the correct Lolyda.AA3 example, the methods emphasize localized texture structure rather than the relatively uniform regions. The explanations are not identical because each method operates on a different representation, but they provide complementary evidence about which byte-image areas influence the prediction.

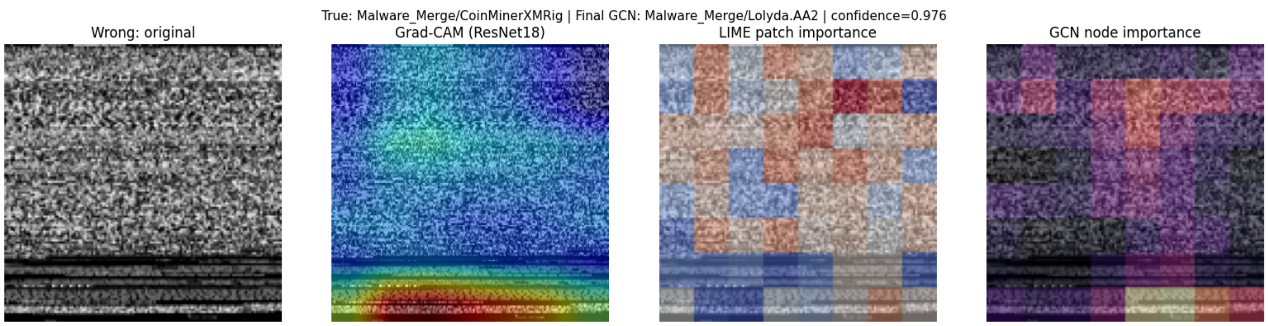

Figure 9.2: Wrong prediction - CoinMinerXMRig classified as Lolyda.AA2 (GCN confidence 0.9759)

For the misclassified CoinMinerXMRig sample, importance is distributed across several central and lower patches. The high confidence combined with an incorrect class suggests that some texture patterns are shared with another family and are not sufficiently specific to the true class.

MaleBin images are byte-derived grayscale representations rather than natural scenes. Highlighted regions therefore indicate model-sensitive image areas and should not be interpreted directly as specific malware functions, code blocks, or causal program behavior.

10. Comparison with Updated Track 2 Related Work

The related-work set was revised to match Track 2. All five studies below explicitly convert images into graphs or apply graph-based image classification. Because the studies use different datasets and experimental protocols, they are used for methodological comparison rather than direct numerical benchmarking against MaleBin.

Study	Dataset	Graph representation	GCN/GNN type	Relation to this project
Vision GNN: An Image is Worth Graph of Nodes (2022)	ImageNet-1K	Fixed image patches + nearest-neighbor graph	ViG / graph convolution	Supports patch-node image graphs
Superpixel Image Classification with Learnable Positional Embedding (2022)	FashionMNIST, CIFAR-10, SVHN, CAR196, CUB200, EuroSAT	SLIC superpixels + region adjacency + position	ChebNet / GCN / GAT	Shows value of spatial/positional information
Image Classification using GNN and Multiscale Wavelet Superpixels (2023)	MNIST, FashionMNIST, CIFAR-10	WaveMesh multiscale superpixels	SplineCNN	Shows sensitivity to graph/pooling design
Geometric Superpixel Representations for Efficient Image Classification with GNNs (2023)	CIFAR-10, ImageNet-1K	Region-adjacency superpixel graph	ShapeGNN	Highlights richer node geometry/features

Enhanced Image Classification with Superpixel-Derived CNN Features (2025)	Agricultural image datasets	Superpixels with CNN-derived node features	GCN graph classifier	Directly motivates CNN-feature nodes used in Task 3

The updated literature reinforces the Task 3 design choice. ViG demonstrates patch-based image graphs, while the superpixel studies show that graph construction and positional structure strongly affect performance. Most importantly, the 2025 hybrid study uses CNN-derived visual features as graph-node attributes, which closely parallels the improvement made here from handcrafted patch statistics to learned CNN node features. The main difference is that MaleBin byte-images do not contain naturally meaningful object regions, so this project uses a regular spatial patch graph and evaluates its usefulness through ablation rather than assuming semantic superpixels.

11. Reproducibility and Limitations

The selected Task 3 architecture is CNN_GCN_4N. The experiments use a leakage-controlled split and save the final model configuration, label map, and trained checkpoints. The saved Kaggle version may show small numerical variation from an interactive training run because GPU training is stochastic, but the selected architecture and overall conclusions remain unchanged.

The five-fold evaluation uses only three training epochs per fold because of computational limits. Longer training and wider hyperparameter search could change absolute scores. In addition, the regular patch graph is a practical spatial construction for byte-images, but it is not guaranteed to provide the same semantic advantages that superpixel graphs provide in natural-image tasks.

12. Conclusion

Task 3 substantially improved the graph-based malware classifier. Replacing six handcrafted patch statistics with learned CNN feature nodes increased macro-F1 from 0.3771 for the Task 2 first PatchGCN to 0.7857 for the selected 4-neighbor CNN-feature GCN. Ablation showed that graph message passing improved the CNN-only backbone, while denser weighted 8-neighbor connectivity did not improve macro-F1 and batch normalization was beneficial. ResNet18 remained stronger on the fixed test split, and McNemar testing confirmed that paired difference, while the five-fold Wilcoxon test did not establish a significant fold-wise difference. The final GCN is therefore best interpreted as a much stronger and compact graph-based alternative rather than a replacement for the strongest CNN baseline.

References

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